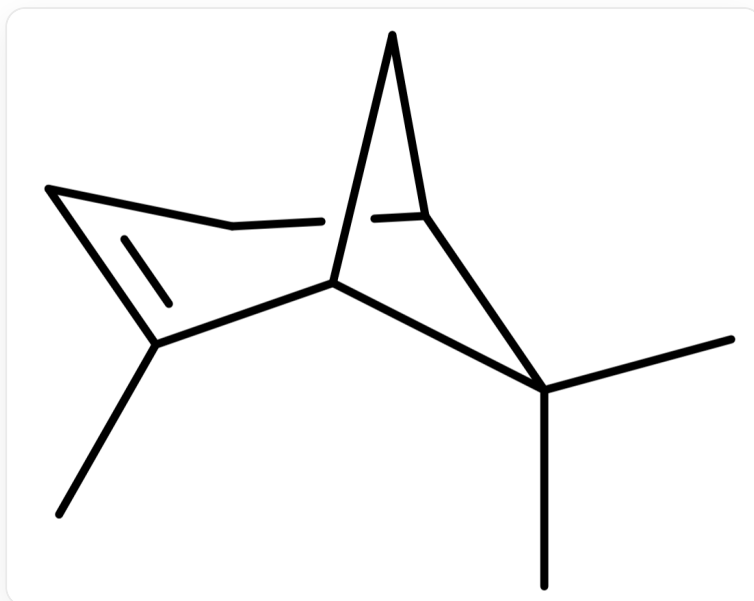


Question

The structure of (-)- α -pinene is shown in the figure.



The specific rotation of a solution of (-)- α -pinene is -50.7° . (-)- α -Pinene is mixed with its enantiomer in a certain ratio, and the specific rotation of the resulting solution is -12° . Calculate the enantiomeric excess e.e. of the mixture.

(-)- α -Pinene is added to 5% H_2SO_4 to generate an optically inactive product **A**.

At low temperature, (-)- α -pinene reacts with one equivalent of HCl in diethyl ether to produce **B**₁ and **B**₂. If the reaction is carried out at room temperature, the products are **B**₃ and a small amount of **B**₄. The formation of **B**₁, **B**₂, **B**₃, **B**₄ all proceed through a single intermediate. **B**₁ and **B**₂ are stereoisomers, and **B**₃ and **B**₄ are constitutional isomers. The rate of elimination reaction with sodium ethoxide: **B**₂ > **B**₁.

(-)- α -Pinene reacts with NOCl , and then treated with KOH to generate compound **C**. **C** is treated with $\text{Zn}/\text{CH}_3\text{COOH}$ to obtain stereoisomers **D**₁ and **D**₂. The rate of reaction with 2-chloropentane: **D**₁ > **D**₂.

Which of the following options is correct:

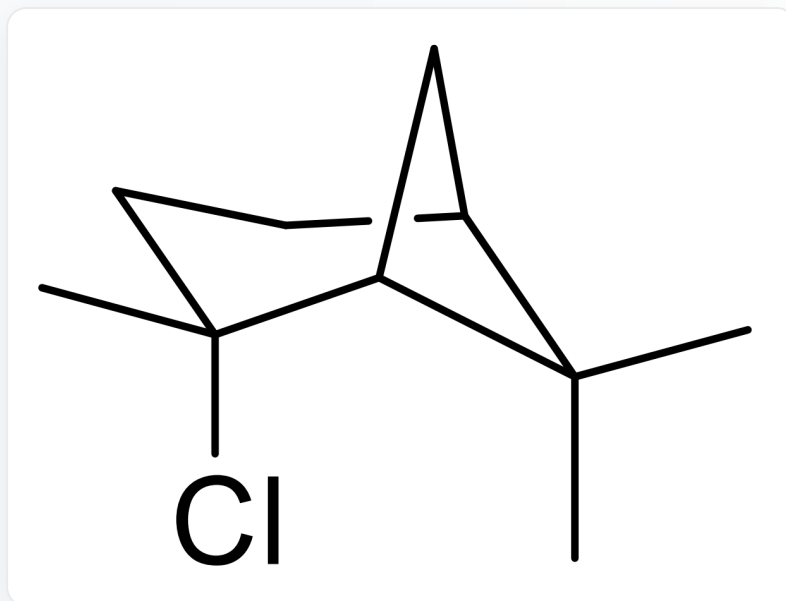
A. All options other than A and B are incorrect.

B. Other than A and B, more than 1 option is correct.

C. The e.e. value of the mixture is greater than 25%

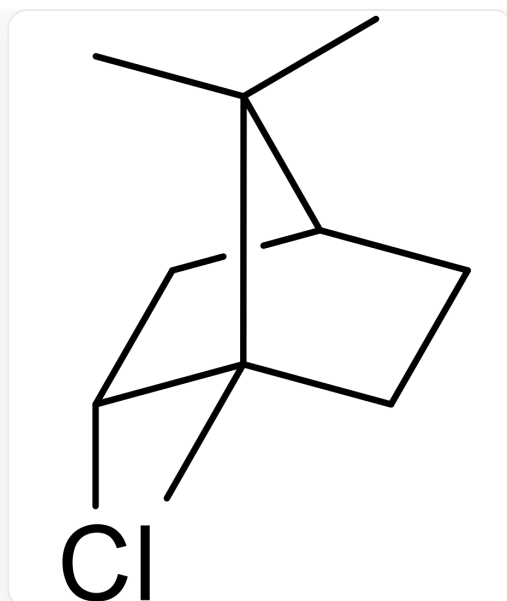
D. The molecular weight of **A** is less than 170.

E. The structure of **B₁** is



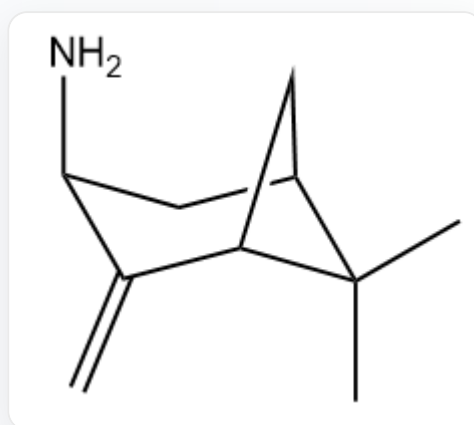
C1C[C@](Cl)(C)[C@@]2(C[C@@]1([H])C2(C)C)[H]

F. **B₄**'s structure is



C1[C@@H](Cl)[C@]2(C)C(C)(C)[C@H]1CC2

G. The structure of D_1 is



C1[C@H](N)C(=C)[C@@H]2C[C@H]1C2(C)C

Answer

Correct Answer: **A**

Detailed Explanation

Mixture's enantiomeric excess e.e. = $|\alpha|_{mix}/|\alpha|_{pure} \times 100\% = 12/50.7 = 23.7\%$.

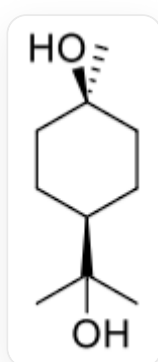
It can be seen that e.e. is no greater than 25%. Therefore, option C is incorrect.

CHECKPOINT

0.5 PTS

e.e. = 23.7%, option C is incorrect

Under 5% H_2SO_4 conditions, α -pinene is protonated to obtain a carbocation, undergoing ring-opening and addition to obtain **A** as shown in the figure below:



CC(O)([C@H]1CC[C@@H](O)CC1)C

The molecular formula is $\text{C}_{10}\text{H}_{20}\text{O}_2$, and the molecular weight is 172.26, option D is incorrect.

CHECKPOINT

1 PTS

A's molecular formula is $C_{10}H_{20}O_2$, molecular weight is 172.26, greater than 170, option D is incorrect

CHECKPOINT

1 PTS

A's structure is CC(O)([C@H]1CC[C@@H](O)CC1)

The classical pinene-hydrochloric acid reaction proceeds through a unique non-classical carbocation intermediate. At low temperatures, the chloride anion is rapidly intercepted, yielding a pair of stereoisomeric chlorides **B**₁, **B**₂. At higher temperatures, it is further converted to structurally isomeric **B**₃, **B**₄ via Wagner–Meerwein rearrangement (commonly described as bornyl chloride **B**₃ and a small amount of fenchyl chloride **B**₄ in biology), which are structural isomers of each other.

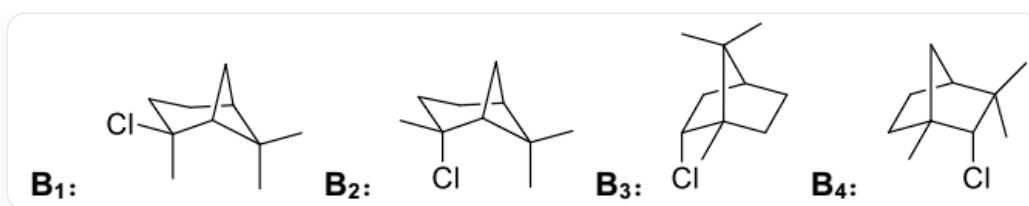
The elimination rates (sodium ethoxide) of **B**₁ and **B**₂ are different, **B**₂ > **B**₁, reflecting the difference in the accessibility of their trans-elimination (trans-coplanar geometry), which is consistent with the two stereoisomers (same skeleton) obtained at low temperatures.

CHECKPOINT

1 PTS

B₁ undergoes E2 elimination reaction slower than **B**₂, indicating that **B**₁ is a trans structure, option E is incorrect

The **B**₄ structure given in option F does not match the corresponding relationship between the skeleton and the substitution position of the classical minor product (anisole/anisyl chloride), which is incorrect.



B₁: C1C[C@@](Cl)(C)[C@@]2(C[C@@]1([H])C2(C)C)[H]; **B₂:** C1C[C@](Cl)(C)[C@@]2(C[C@@]1([H])C2(C)C)[H]; **B₃:** C1[C@@H](Cl)[C@]2(C)C(C)(C)[C@H]1CC2; **B₄:** C1C[C@@]2(C[C@H]1C(C)(C)[C@@H]2Cl)C

CHECKPOINT

1 PTS

During this rearrangement reaction, the migration rate of the quaternary carbon is faster, and the option's structure is the major product **B₃**, option F is incorrect

CHECKPOINT

1 PTS

B₁'s structure is C1C[C@@](Cl)(C)[C@@]2(C[C@@]1([H])C2(C)C)[H]

CHECKPOINT

1 PTS

B₂'s structure is C1C[C@](Cl)(C)[C@@]2(C[C@@]1([H])C2(C)C)[H]

CHECKPOINT

1 PTS

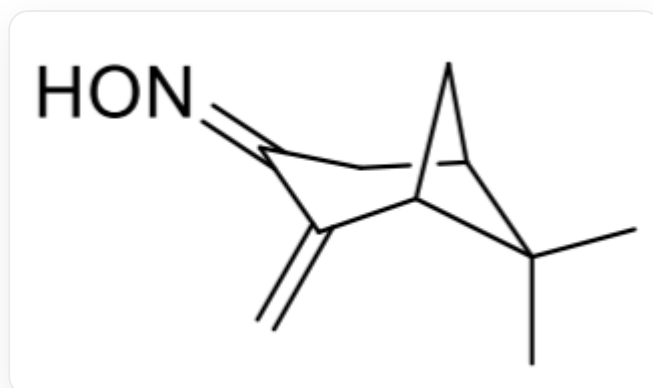
B₃'s structure is C1[C@@H](Cl)[C@]2(C)C(C)(C)[C@H]1CC2

CHECKPOINT

1 PTS

B₄'s structure is C1C[C@@]2(C[C@H]1C(C)(C)[C@@H]2Cl)C

The addition of α -pinene and NOCl yields a chlorinated hydrocarbon containing a nitroso group. Under alkaline conditions, the nitroso group tautomerizes to an oxime, and then an elimination reaction occurs to obtain **C**;



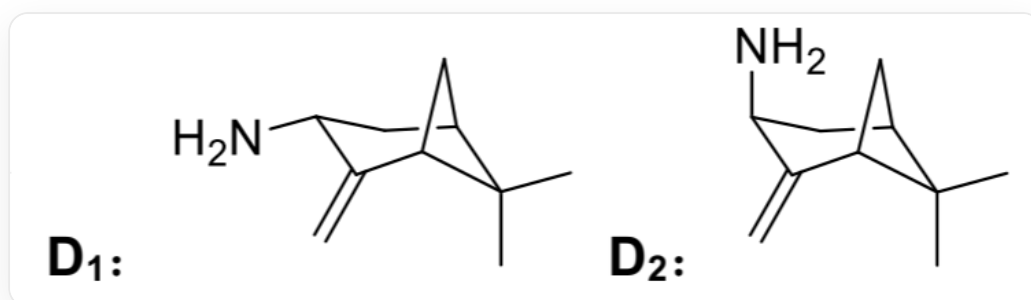
C结构为C1C(=NO)C(=C)[C@]2([H])C[C@@]1([H])C2(C)C

CHECKPOINT

1 PTS

C's structure is C1C(=NO)C(=C)[C@]2([H])C[C@@]1([H])C2(C)C

Zn/AcOH reduction yields the corresponding amine. The generated amine has two stereoisomers (exo/endo), where the exo-substituted amine has less steric hindrance and exhibits a more pronounced nucleophilicity, resulting in a faster substitution reaction (quaternization) with secondary haloalkanes (such as 2-chloropentane), so $D_1 > D_2$.



D_1 结构为 C1[C@@]2(C(C)(C)[C@](C2)([H])C(=C)[C@@H]1N)[H], D_2 结构为 C1[C@@]2(C(C)(C)[C@](C2)([H])C(=C)[C@H]1N)[H]

The D_1 structure given in option G is the endo-camphornene amine configuration, which does not match the rate comparison. Therefore, G is incorrect.

CHECKPOINT

1 PTS

The exo-substituted amine has less steric hindrance and exhibits a more pronounced nucleophilicity, resulting in a faster substitution reaction with secondary haloalkanes

CHECKPOINT

1 PTS

D_1 's structure is C1[C@@]2(C(C)(C)[C@](C2)([H])C(=C)[C@@H]1N)[H]

CHECKPOINT

1 PTS

D₂'s structure is C1[C@@]2(C(C)(C)[C@](C2)([H])C(=C)[C@H]1N)[H]

In summary: C, D, E, F, and G are all incorrect, so A should be selected.